

Master's Thesis
**Multi-objective Optimisation of Suction Cup and
Parallel Jaw Grippers**

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Zitat

‘Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.’

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

‘Aushängeschild’ der Arbeit, max 1 Seite

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Nomenclature

(x_i, y_i) Planar position of the i -th suction cup

θ Algorithm hyperparameters

δ_{intra} Spatial threshold for selecting additional local representatives

ℓ_i Lower bound of the i -th decision variable

Λ Set of reference directions for decomposition-based algorithms

$\mathbf{f}(\mathbf{p})$ Multi-objective function vector

$\mathbf{p}$ Decision variable vector representing the gripper configuration

$\mathbf{r}$ Reference point for hypervolume computation

$\mathcal{A}$ Selected multi-objective evolutionary algorithm

$\mathcal{F}^{(t)}$ Objective values of the population at generation t

$\mathcal{G}_{\text{cand}}$ Set of candidate grasps generated from sampled surface points

$\mathcal{G}_{\text{rep}}$ Reduced representative grasp set used for diversity evaluation

$\mathcal{G}_{\text{valid}}$	Set of valid (collision-free) grasps
$\mathcal{P}^*$	Set of Pareto-optimal solutions
$\mathcal{P}^{(0)}$	Initial population of candidate solutions
$\mathcal{S}$	Set of sampled surface points on the workpiece
$\mathcal{Z}$	Set of random seeds for meta-optimization
$\text{HV}^{(t)}$	Hypervolume indicator at generation t
$A(\mathbf{p})$	Area of the triangle formed by the three suction cups
$D(\mathbf{p})$	Diversity metric of the grasp distribution
$d_{\text{geo}}(\mathbf{R}_a, \mathbf{R}_b)$	Geodesic distance between two grasp orientations
$d_{\text{ori}}(i, j)$	Pairwise orientation distance between grasp poses i and j
$D_{\text{pos}}(\mathbf{p})$	Spatial component of the diversity metric
$d_{\text{pos}}(i, j)$	Pairwise translational distance between grasp poses i and j
$D_{\text{rot}}(\mathbf{p})$	Orientation component of the diversity metric
E_{max}	Maximum number of function evaluations
f_{tol}	Objective space convergence tolerance
G_{max}	Maximum number of generations
M	Number of initially sampled surface points

N	Population size
n	Number of decision variables
n_{obj}	Number of Objectives
q_{intra}	Maximum number of local orientation-diverse representatives
s	Characteristic object scale derived from the mesh bounding box
t	Generation index
u_i	Upper bound of the i -th decision variable
x_{tol}	Design space convergence tolerance

1 Introduction

1.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics[5]. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments[6]. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces[7].

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches[8]. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. Designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints[9].

While the gripper configuration problems are well-motivated in themselves, the primary objective of this thesis is not simply to find the best gripper configuration, but to systematically apply and compare optimization algorithms for solving these configuration problems. A large number of optimization algorithms exist, spanning evolutionary strategies, decomposition-based methods, indicator-based selection, and Bayesian optimization, yet their relative suitability for gripper design problems of this type is not well understood. Different algorithms may exhibit substantially different convergence behaviour, solution quality, and robustness depending on the structure of the objective landscape and the dimensionality of the search space. Identifying which algorithm class performs best for each gripper problem is therefore a practically relevant and scientifically meaningful research question[10].

1.2 Problem Definition

Designing grippers for a given set of workpieces is still largely a manual, experience-driven task, in which geometric parameters such as suction-cup placement or finger contours are chosen iteratively and validated in simulation or on hardware[11]. Because grasp quality can only be assessed through a simulation or geometric evaluation routine, no analytical gradients or closed-form objective expressions are available, and each evaluation is comparatively expensive[12]. Moreover, a single gripper usually has to handle several workpieces at once, so the design criteria are inherently conflicting and cannot be reduced to a single scalar quality measure without losing information about the available trade-offs. These properties of expensive, derivative-free, multi-objective evaluations place gripper design squarely in the domain of black-box multi-objective optimization[13]. Which class of optimization algorithm is best suited for such problems, however, remains an open question, as performance is known to depend strongly on the number of objectives, the shape of the objective landscape, and the available evaluation budget[12, 14].

The primary goal of this thesis is to apply and comparatively evaluate optimization algorithms spanning evolutionary multi-objective methods and Bayesian optimization for solving gripper configuration problems formulated as black-box multi-objective optimization tasks[15, 16]. The two gripper types, a multi-suction-cup gripper and a two-finger gripper, serve as well-defined benchmark problems on which the algorithms are systematically compared with respect to solution quality, convergence behaviour, and robustness.

To provide a rigorous and reproducible basis for this comparison, two gripper configuration problems are formulated as well-defined multi-objective optimization tasks. In both cases, a diverse set of workpieces is incorporated directly into the problem formulation by treating each workpiece as a separate objective. This design choice ensures that the optimized gripper configurations are not tailored to a single geometry, but are instead robust across a range of workpiece shapes. It also increases the dimensionality of the objective space as the number of workpieces grows, allowing the benchmarking study to probe algorithm behaviour under varying levels of problem complexity[17].

For the **multi-suction-cup gripper**, the objectives consist of the geometric stability of the suction base together with the grasp diversity evaluated independently for each workpiece. These objectives are inherently conflicting: a larger suction base improves stability but increases the risk of collisions with the workpiece, reducing the number of feasible grasps.

For the **two-finger gripper**, a fixed number of grasp points is defined for each workpiece, and the Objectives are computed from the conformance of the finger contour to the corresponding workpiece section at each grasp point. A finger contour that scores well at all grasp points simultaneously must therefore generalize across diverse geometric profiles, which makes the problem increasingly challenging as the number of grasp points grows.

Together, these two problems provide a structured experimental basis for comparing optimization algorithms across varying objective space dimensionalities and objective landscape characteristics, within the practically relevant domain of robotic gripper design.

To achieve this goal, the following tasks are defined:

- **Problem formulation:** Formulation of both gripper configuration problems as multi-objective optimization problems, including the definition of decision variables, objective functions, and constraints.
- **Optimization framework implementation:** Implementation of a generalized optimization framework capable of handling multi-objective problems of varying dimensionality and interfacing with both evaluation pipelines through a unified black-box interface.
- **Evaluation pipeline development:** Development of dedicated black-box evaluation pipelines for each gripper type, capable of assessing grasp quality given a gripper configuration and a target workpiece, and of serving as the objective function interface for all optimization algorithms.
- **Workpiece selection and experimental coverage:** Selection of a diverse set of workpieces spanning a range of geometric properties including varying surface curvatures, sizes, and shapes to ensure that algorithm comparisons are not biased towards a specific object geometry. Workpiece diversity is treated as a deliberate experimental design choice to expose differences in algorithm robustness and generalization across objective landscapes of varying complexity.
- **Algorithm application and comparison:** Application and systematic comparison of a selected set of multi-objective optimization algorithms drawn from the available literature [18, 3, 19], evaluated under controlled experimental conditions using well-defined performance indicators across both gripper types and workpieces[20].

- **Result analysis:** Analysis and interpretation of algorithm performance across both gripper types and all workpieces, with particular focus on convergence behaviour, Pareto front quality, and the sensitivity of algorithm rankings to workpiece geometry and objective space dimensionality.

2 Literature Review

2.1 Theoretical Background

2.1.1 Robotics and Object Handling

Robotics

An industrial robot is a programmable, multipurpose manipulator designed to move material, parts, tools or specialised devices through programmed motions in order to perform a variety of tasks[21]. A typical robotic system consists of a kinematic chain of rigid links connected by actuated joints, a set of sensors, a controller and an end-effector mounted at the flange of the last link. The configuration of the manipulator is described by its joint variables, while the position and orientation of the end-effector are described in a Cartesian task space[22]. The mapping between these two spaces is established through forward and inverse kinematics, which form the basis for trajectory planning and motion control[23].

Modern industrial applications place increasing demands on flexibility. Robots are expected to operate in environments with varying product portfolios, short product life cycles and small batch sizes, which shifts the engineering effort from robot programming towards the design of task-specific peripheral components[24, 25]. Among these components the end effector is the most application-dependent element, since it forms the physical interface between the robot and the workpiece and therefore determines whether a given handling task can be executed reliably at all[24].

Bin Picking Problem

Bin picking describes the task of removing individual workpieces from an unordered accumulation of parts, typically supplied in a container, and transferring them to a defined target pose[26, 27]. The task combines perception, grasp planning, collision-free motion planning and physical grasp execution. Because the parts are randomly oriented and may be partially occluded or entangled, the pose of the next object to be grasped is not known in advance and must be estimated from sensor data at runtime[28, 29].

For gripper design, the decisive consequence of this setting is that a grasp cannot be planned for one privileged object pose[26]. The gripper must be able to reach and hold the workpiece across a broad range of orientations and partial accessibility conditions, which implies that a useful gripper configuration provides several alternative grasp candidates distributed over the object surface rather than one single optimal contact[26, 30]. Additional constraints arise from the confined working space inside the bin, where collisions between the gripper body and the bin walls or neighbouring parts restrict the set of feasible approach directions[29, 31]. These requirements directly motivate the objectives used in this thesis, in particular the diversity of the available grasps for the suction cup gripper and the geometric conformance of the finger contour for the two-finger gripper.

2.1.2 Gripper Theory and Geometric Representation

Grasp Theory

A grasp is a configuration of contact points between a gripper and a workpiece such that the gripper can resist external forces and torques applied to the object[32]. The classical criterion for grasp quality is *force closure*. A grasp is force-closed if the contact forces available at the contact points can produce any net wrench on the object, meaning that arbitrary disturbances can be resisted[33]. Force closure requires a sufficient number of contacts with appropriate friction properties and geometric placement.

Suction Cup Grippers

Suction cup grippers belong to the astrictive gripper class and generate holding force from a pressure difference between the evacuated volume inside the cup and the ambient atmosphere. In the idealised case the holding force is

$$F_{\text{hold}} = \frac{\Delta p \cdot A}{S} \quad (2.1)$$

where Δp is the pressure difference, A is the effective sealed area and S is a safety factor accounting for acceleration, orientation and friction conditions[34, 35]. The decisive requirement is therefore an airtight seal between the cup lip and the workpiece surface, which restricts applicable surfaces to regions that are sufficiently smooth, non-porous and locally flat or only moderately curved relative to the compliance of the cup.

Practical implementations frequently use several cups arranged on a common mounting plate. Distributing the load over multiple cups increases the total holding force, provides redundancy if individual cups fail to seal, and, most importantly for handling, generates a resistance to tilting moments that a single cup cannot provide[36]. The moment capacity depends on the spatial arrangement of the active cups, which is commonly characterised through the area of the polygon or triangle spanned by the sealing contacts[35]. A large spanned area yields a stable grasp, whereas narrowly clustered cups behave similarly to a single contact. The placement of the cups on the mounting plate is thus the central design variable, and it must be chosen such that valid sealing contacts are achievable on every workpiece of the considered family.

Grasp Diversity

The *diversity* of a grasp set can be quantified by combining two complementary measures of spatial spread.

Geodesic Distance as a Diversity Measure. Orientational diversity is measured by calculating the pairwise geodesic distances between grasp orientations represented by rotation matrices. The geodesic distance describes the minimum angular rotation required to transform one orientation into another, accounting for the geometry of the rotation space. The mean of these pairwise angular distances is normalized by the maximum possible angular distance of 180° , resulting in a diversity score between zero and one. A score close to zero indicates that the grasp orientations are similar, whereas larger values indicate greater average angular separation and thus higher orientational diversity. This measure enables the assessment of how broadly candidate grasps cover different orientations, supporting the selection of diverse grasp configurations[37, 38].

Spatial Diversity via Mean Pairwise Euclidean Distance. The spatial spread of a set of grasps can be quantified using the mean pairwise Euclidean distance between grasp positions in three-dimensional space. To ensure comparability across objects of different sizes, this distance is normalized by a characteristic object scale s . We define s as the Euclidean norm of the axis-aligned bounding-box side lengths:

$$s = \|\mathbf{e}\|_2, \quad (2.2)$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the side lengths of the bounding box. The normalized spatial diversity is then defined as

$$D_{\text{pos}}(\{\mathbf{y}_i\}_{i=1}^n) = \frac{1}{s} \frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^n \|\mathbf{y}_i - \mathbf{y}_j\|_2, \quad (2.3)$$

where $\mathbf{y}_i \in \mathbb{R}^3$ denotes the position of the i -th grasp contact point, and $n \geq 2$ is the total number of grasps considered. Normalization by s makes the metric scale-invariant, allowing diversity values to be compared across workpieces of different sizes. This measure complements the geodesic measure by capturing absolute spatial separation rather than surface-constrained distance.

As illustrated in Figure 2.1, the Euclidean distance between surface points A and B is the straight-line distance through the ambient space, whereas the geodesic distance follows the shortest path constrained to the object surface. Consequently, the geodesic distance better represents the separation of grasp locations along curved workpiece geometries.

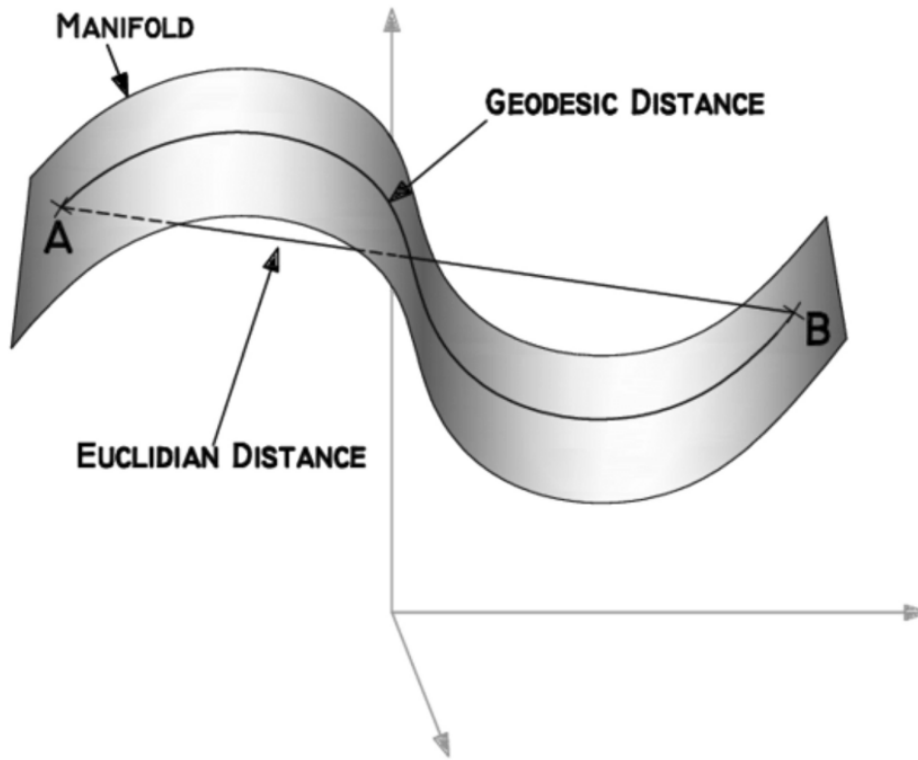


Figure 2.1: Geodesic versus Euclidean distance between grasp locations on an object surface.[1]

Multi Finger Grippers

Multi finger grippers are impactive grippers that transmit forces through direct contact between movable jaws and the workpiece. They are classified by the number of fingers, the kinematics of the jaw motion and the number of actuated degrees of freedom[39]. Two-finger parallel grippers with a single actuator represent the industrial standard because of their mechanical simplicity, robustness and low control effort[40]. Three-finger centric designs provide self-centring behaviour for rotationally symmetric parts, while anthropomorphic hands with many degrees of freedom offer high versatility at the cost of complexity.

Two grasp modes are commonly distinguished. Force-closure grasps hold the object through friction at the contact surfaces and therefore depend on the applied clamping force and the friction coefficient. Form-closure grasps immobilise the object through the geometry of the contacts alone, so that no disturbance can move the object without deformation[41]. Because a mechanically simple gripper provides few actuated degrees of freedom, the ability

to realise form closure is shifted from the kinematics to the geometry of the jaws. This is the reason why application-specific jaw inserts, frequently produced additively, are widely used in industry and why the jaw contour is the natural design variable for optimization[42].

Two-Finger Gripper and Contour Representation

The two-finger gripper applies gripping forces through two opposing jaw surfaces whose contour geometry determines how well the gripper conforms to the workpiece. Designing the jaw contour requires a representation that is compact enough to serve as a low-dimensional decision space for optimization, yet expressive enough to capture smooth, physically realizable shapes[9]. Spline parameterization is one of the approaches for this purpose.

Cubic Spline Interpolation. A cubic interpolating spline is a piecewise polynomial curve constructed to pass through a given set of data points $\{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_m\}$. Each segment is represented by a polynomial of degree at most three, with adjacent segments joined such that the curve and its first and second derivatives are continuous. The resulting curve is therefore smooth while interpolating all prescribed points exactly. To obtain a unique spline, appropriate boundary conditions must be specified at the endpoints.

Spline-Based Shape Parameterization. Spline-based shape parameterization represents a geometric contour through a small set of control points, reducing an otherwise infinite-dimensional shape space to a finite-dimensional parameter vector. Boundary control points are typically fixed to satisfy geometric constraints, while interior control points are free to vary and serve as the design variables. The number of free control points governs the trade-off between geometric expressiveness and search space dimensionality, since more control points allow more complex shapes but increase the dimensionality of the optimization problem[43, 44].

Figure 2.2 illustrates a cubic interpolating spline passing through its prescribed data points. In contrast to a B-spline control-point representation, the interpolation points lie directly on the resulting curve. Consequently, the three variable interior points provide intuitive geometric parameters for modifying the gripper contour, while the fixed endpoints preserve its prescribed spatial extent.

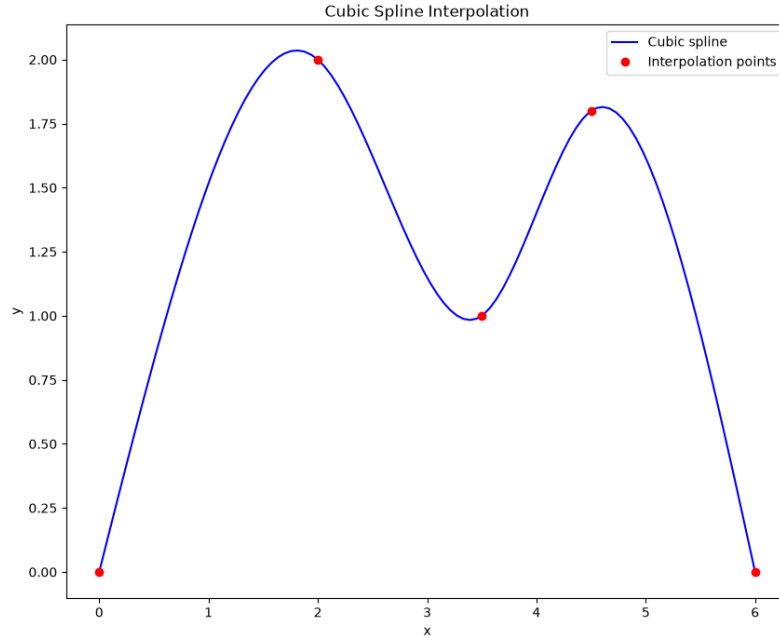


Figure 2.2: Cubic spline interpolation through prescribed data points.

2.1.3 Optimization Theory

Black-Box Optimization

A black-box optimization problem is one in which the objective function has no known analytical form and no gradient information is available to the optimizer[45]. The function can only be queried at candidate input points, returning a scalar or vector output. This setting arises naturally in engineering design when objectives are evaluated through computationally expensive procedures, making gradient computation either impossible or impractical. Because each function evaluation may be costly, black-box optimizers must find high-quality solutions within a limited evaluation budget, placing emphasis on sample efficiency[46]. The gripper configuration problems addressed in this thesis fall squarely within this category. Objective values are computed through a series of geometric and numerical calculations applied to the candidate gripper configuration, and no closed-form expression exists that directly maps the decision variables to either the contact quality score or the grasp diversity

metric.

Multi-Objective Optimization

Multi-objective optimization (MOO) refers to the simultaneous optimization of two or more conflicting objective functions over a shared decision space[47]. Formally, a multi-objective minimization problem is defined as

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})) \quad (2.4)$$

where $\mathbf{x}$ is the decision variable vector, $\mathcal{X}$ is the feasible decision space, and $k \geq 2$ is the number of objectives. When objectives conflict, no single solution minimizes all objectives simultaneously. Instead, a set of trade-off solutions exists, collectively forming the Pareto front. The goal of a multi-objective optimizer is therefore not to find a single best solution but to approximate this trade-off surface as accurately and completely as possible[47, 48].

Pareto Optimality

The concept of Pareto optimality provides the theoretical foundation for comparing solutions in a multi-objective setting. A solution $\mathbf{x}^{(a)}$ is said to *dominate* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$, if and only if

$$f_i(\mathbf{x}^{(a)}) \leq f_i(\mathbf{x}^{(b)}) \quad \forall i \in \{1, \dots, k\} \quad \text{and} \quad f_j(\mathbf{x}^{(a)}) < f_j(\mathbf{x}^{(b)}) \quad \text{for at least one } j \quad (2.5)$$

A solution is called *Pareto optimal* if no other feasible solution dominates it. The set of all Pareto-optimal solutions in the decision space is the *Pareto set*, and its image in the objective space is the *Pareto front*. In practice, algorithms compute an approximation of the Pareto front, and the quality of this approximation is assessed using performance indicators such as the hypervolume indicator[20, 49].

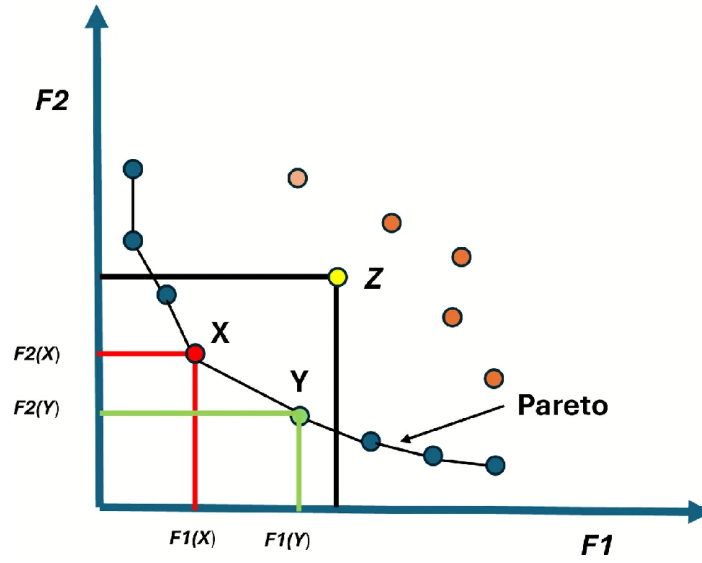


Figure 2.3: Example of a Pareto Front.[2]

Figure 2.3 illustrates the concept of dominated and non-dominated solutions in a multi-objective optimization (MOP) problem with two objectives. The objective is to minimize both objective functions. The connected curve represents the Pareto front (PF), which includes all Pareto-optimal solutions. Solution X_2 has higher values for both f_1 and f_2 when compared to solutions X_1 and Y . Thus, solutions X_1 and Y dominate solution X_2 . However, since neither X_1 nor Y dominates the other, they are considered non-dominated solutions[2].

Many-Objective Optimization

Problems with four or more objectives are commonly referred to as many-objective optimization problems[50]. As the number of objectives grows, the proportion of mutually non-dominated solutions in a randomly sampled population increases rapidly, an effect known as dominance resistance. Pareto dominance then loses its ability to discriminate between candidates, so selection pressure towards the Pareto front deteriorates and the search degenerates into an essentially random walk guided mainly by diversity preservation[50, 51].

The number of solutions required to cover a k -dimensional front grows exponentially with k , visual inspection of the approximation set becomes impractical, and the computation of indicators such as the hypervolume becomes expensive[4]. Algorithm design responds to

these issues either by replacing dominance with decomposition into scalar subproblems, by introducing reference points or reference directions to structure the objective space, or by using an indicator as the selection criterion[50, 52].

No Free Lunch Theory

The No Free Lunch (NFL) theorems for optimization state that, averaged over the set of all possible objective functions, all non-repeating search algorithms exhibit identical expected performance[53, 54]. Any advantage that an algorithm demonstrates on one class of problems is therefore compensated by a corresponding disadvantage on another class. Corresponding results have been derived for the multi-objective case[54].

The practical consequence is that there is no universally superior optimization algorithm and that the question of which algorithm to use is only meaningful with respect to a specific problem class[55]. This provides the methodological justification for the empirical approach taken in this thesis. Rather than seeking a generally best algorithm, the objective is to determine which algorithms perform well on the particular structure of gripper configuration problems, and how sensitive this ranking is to workpiece geometry and to the dimensionality of the objective space.

Evolutionary Algorithms

Evolutionary algorithms (EAs) are a family of population-based metaheuristics inspired by the principles of biological evolution. A population of candidate solutions is maintained across discrete generations. At each generation, new candidate solutions are generated through variation operators, typically crossover, which recombines pairs of existing solutions, and mutation, which introduces random perturbations, and a selection mechanism retains the most promising individuals for the next generation based on their objective values. This loop continues until a termination criterion is met, such as a maximum number of function evaluations or convergence of the population[56].

EAs are well suited to black-box multi-objective optimization because they require no gradient information, naturally maintain a diverse population of trade-off solutions, and are robust to noisy or discontinuous objective landscapes. Multi-objective evolutionary algorithms (MOEAs) extend the basic EA framework with selection mechanisms based on Pareto

dominance or decomposition, enabling them to approximate the Pareto front in a single run[57, 58].

Bayesian Optimization

Bayesian optimization (BO) is a sequential model-based strategy for the global optimization of expensive black-box functions. A probabilistic surrogate model, most commonly a Gaussian process (GP), is fitted to the observations collected so far[59]. The surrogate provides both a mean prediction and an uncertainty estimate at unobserved locations, which together define an *acquisition function* that balances exploration of uncertain regions with exploitation of promising ones[59, 60]. The point that maximizes the acquisition function is selected as the next evaluation, after which the surrogate is updated and the process repeats[61][3]. Because the surrogate concentrates the sampling effort on informative regions, BO typically requires far fewer function evaluations than population-based methods, which makes it attractive whenever the evaluation budget is the limiting factor.

Gaussian Process Surrogates. A Gaussian process defines a distribution over functions and is fully specified by a mean function $m(\mathbf{x})$ and a covariance or kernel function $k(\mathbf{x}, \mathbf{x}')$, written as $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$. Conditioning the prior on the observed data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ yields a posterior that is again Gaussian, with closed-form predictive mean $\mu(\mathbf{x})$ and predictive variance $\sigma^2(\mathbf{x})$ at any query point[62]. The kernel encodes the assumed smoothness and length scale of the underlying function, with the Matérn and squared exponential kernels being the most common choices, and its hyperparameters are estimated by maximizing the marginal likelihood[62]. An additional noise term accounts for stochastic or numerically imprecise evaluations[63]. The main limitations of GP surrogates are the cubic scaling of exact inference with the number of observations and a loss of predictive quality in high-dimensional decision spaces[59].

In figure 2.4 a surrogate model is fit to five noisy observations using a GP to predict the objective, depicted by the solid line, and uncertainty estimates, illustrated by the width of the shaded bands. This objective is predicted for the entire range of possible parameter values, corresponding to the full x-axis. Importantly, the model is able to predict the outcome and quantify the uncertainty of configurations that have not yet been tested. Intuitively, the

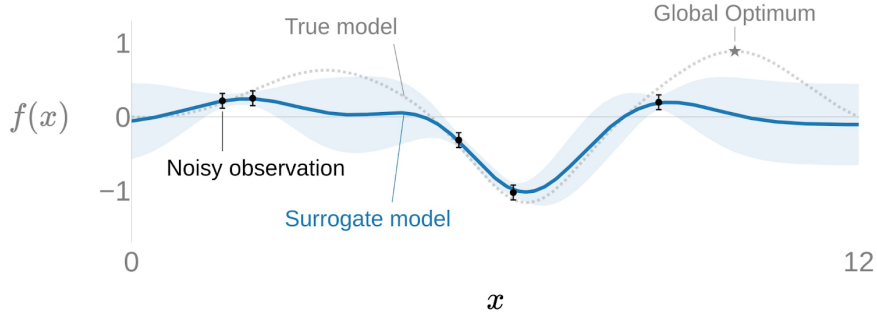


Figure 2.4: Surrogate Model [3].

uncertainty bands are tight in regions that are well-explored and become wider as we move away from them.

Acquisition Functions. The acquisition function converts the posterior of the surrogate into a scalar utility that quantifies how promising a candidate point is. Expected improvement (EI) measures the expected amount by which a candidate improves on the best observed value,

$$\alpha_{\text{EI}}(\mathbf{x}) = \mathbb{E}[\max(f_{\min} - f(\mathbf{x}), 0)] \quad (2.6)$$

and admits a closed-form expression for a Gaussian posterior[63, 3].

A visualization of the expected improvement based on the surrogate model predictions is shown in Figure 2.5, where the next suggestion is where the expected improvement is at its maximum. Once a new highest EI is selected and evaluated, the surrogate model is retrained and a new suggestion is made. This process continues iteratively until a stopping condition, set by the user, is reached [3].

Since the acquisition function is inexpensive to evaluate and analytically differentiable, its maximization is performed by an inner optimizer rather than by querying the expensive objective.

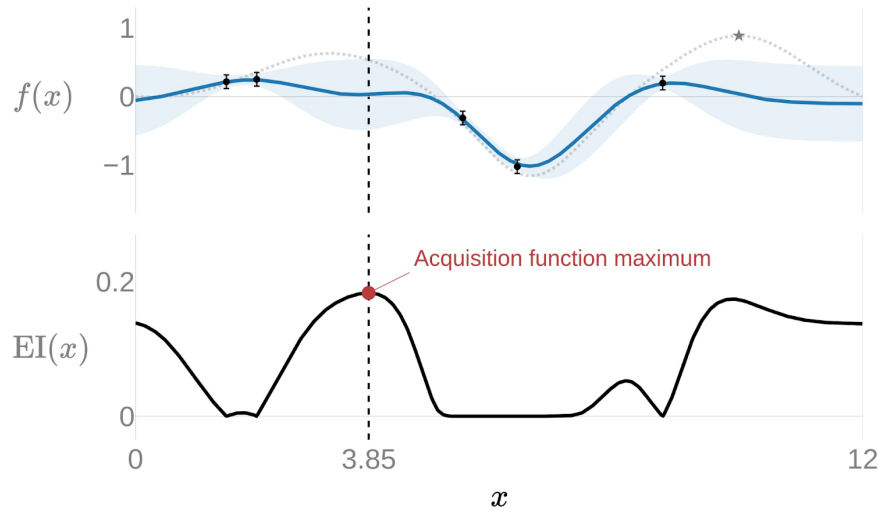


Figure 2.5: Acquisition Function [3].

Multi-Objective Bayesian Optimization. Extending BO to several objectives requires a utility notion for sets instead of single points. One family of methods fits an independent surrogate per objective and scalarizes the problem with randomly drawn weight vectors, so that the front is sampled gradually over successive iterations[64]. A second and now dominant family uses the hypervolume as the underlying utility[65]. Expected hypervolume improvement and its noisy, batch-capable variants select the candidates that are expected to enlarge the dominated hypervolume the most, which aligns the acquisition criterion directly with the indicator used for evaluation[64]. Information-theoretic approaches instead maximize the information gained about the Pareto front. Multi-objective BO is particularly appealing under tight evaluation budgets, but its cost per iteration grows with the number of objectives and observations, which makes the comparison against evolutionary methods on the gripper problems a question of the available budget rather than of asymptotic quality alone.

Performance Indicators

Because a multi-objective optimizer returns a set rather than a single point, comparing algorithms requires indicators that map an approximation set to a scalar quality value[4]. Indicators are usually grouped according to whether they assess convergence towards the front, the diversity of the solutions along the front, or both properties at once.

The *hypervolume* indicator (HV) measures the volume of the objective space that is dominated by the approximation set and bounded by a reference point $\mathbf{r}$,

$$\text{HV}(A, \mathbf{r}) = \lambda\left(\bigcup_{\mathbf{a} \in A} [\mathbf{f}(\mathbf{a}), \mathbf{r}]\right) \quad (2.7)$$

where λ denotes the Lebesgue measure[20, 66]. The hypervolume is the most widely used indicator because it is Pareto compliant, meaning that a strictly dominating set never receives a worse value, and because it requires no knowledge of the true Pareto front[4]. Its drawbacks are the growth of its computational cost with the number of objectives and its sensitivity to the choice of the reference point[4, 20].

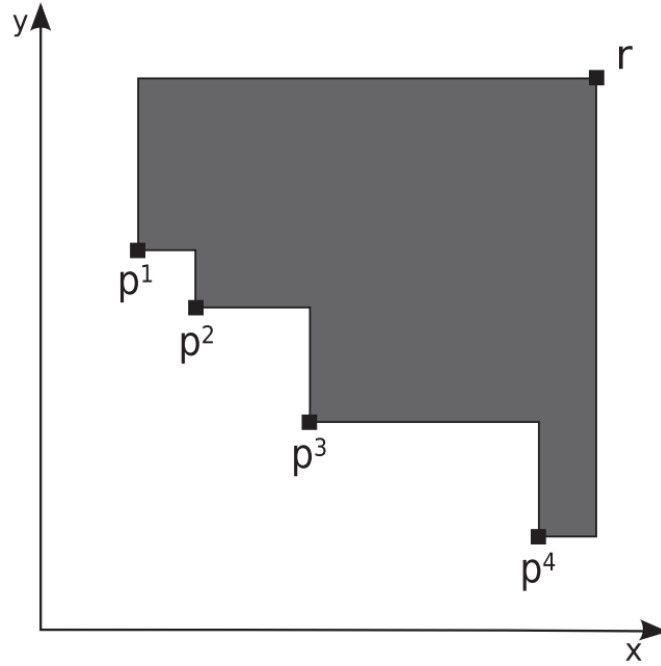


Figure 2.6: Hypervolume indicator [4].

Figure 2.6 shows a two-dimensional example of the hypervolume (an area) and a fixed reference point $\mathbf{r}$.

The *inverted generational distance* (IGD) measures the mean distance from each point of a

reference set to the closest point of the approximation set and therefore requires a reference front, which is normally unavailable for real-world black-box problems[66]. In this thesis the hypervolume serves as the primary indicator, supplemented by convergence histories over the number of function evaluations so that anytime behaviour is captured in addition to final solution quality.

2.1.4 Multi-Objective Evolutionary Algorithms

The candidate algorithms evaluated in this thesis are all members of the multi-objective evolutionary algorithm (MOEA) family. Each follows the general EA loop described in Section 2.1.3 but differs in how it guides selection toward a well-distributed approximation of the Pareto front. Brief introductions to each algorithm are given below.

NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is among the most widely adopted MOEAs. Its selection mechanism is based on two criteria applied in sequence. First, the entire population is sorted into non-domination ranks, so that all non-dominated solutions receive rank 1, they are removed, and the process repeats until every individual has a rank. Second, within the same rank, solutions are ordered by *crowding distance*, a density estimate that measures how isolated a solution is from its neighbours in the objective space. Solutions with a higher crowding distance are preferred, encouraging spread along the Pareto front. NSGA-II uses an elitist strategy in which the combined parent and offspring population of size $2N$ is truncated to N using these two criteria. Its computational complexity per generation is $\mathcal{O}(kN^2)$, where k is the number of objectives and N is the population size[67].

NSGA-III

NSGA-III extends NSGA-II to problems with three or more objectives (many-objective problems), where crowding distance becomes an unreliable diversity measure. Instead of crowding distance, NSGA-III uses a set of structured *reference points* distributed uniformly on a normalized hyperplane in the objective space. Solutions are associated with their closest

reference point, and selection preference is given to solutions that are associated with under-represented reference points and lie closest to their reference line. This mechanism explicitly promotes diversity across the objective space and has been shown to outperform NSGA-II on problems with three or more objectives[68].

MOEA/D

MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) takes a fundamentally different approach from Pareto-ranking methods. The multi-objective problem is decomposed into N scalar subproblems using a set of evenly distributed weight vectors and an aggregation function such as the weighted Chebyshev scalarization. Each subproblem is optimized simultaneously, with solutions sharing information within a defined neighbourhood of weight vectors. This neighbourhood-based collaborative search makes MOEA/D computationally efficient and well suited to problems where the Pareto front has a regular structure. Its complexity per generation is lower than NSGA-II, as it avoids global non-dominated sorting[69, 18].

SMS-EMOA

The S-Metric Selection Evolutionary Multi-Objective Algorithm (SMS-EMOA) uses the hypervolume indicator directly as its selection criterion. It operates as a steady-state algorithm, so that at each step a single offspring is generated, added to the population, and the individual that contributes the least to the population hypervolume is removed. This ensures that the population always converges toward solutions that maximize the dominated hypervolume with respect to a reference point. SMS-EMOA is particularly well suited to problems where solution quality is to be measured by the hypervolume indicator, as its selection pressure is directly aligned with that metric[70].

CMOPSO

The Competitive Multi-Objective Particle Swarm Optimizer (CMOPSO) adapts the particle swarm optimization (PSO) framework to the multi-objective setting. In standard PSO, each particle moves through the decision space guided by its personal best position and a global

best position. CMOPSO introduces a competitive mechanism in which particles compete in pairs within the swarm, and the losing particle updates its velocity by learning from the winning particle rather than from a fixed global best. An external archive of non-dominated solutions is maintained to provide guidance and preserve diversity across the Pareto front. This competitive selection strategy avoids premature convergence to a single region of the Pareto front and has been shown to achieve fast convergence while remaining competitive with dominance-based MOEAs on standard benchmark problems[71].

2.2 State of the Art

This section compares previous studies according to their optimization target, the gripper properties treated as fixed, and their relevance to task-specific gripper configuration. The reviewed literature can be divided into four groups: grasp planning with fixed end-effectors, multi-suction grasping, two-finger gripper design, and comparative studies of multi-objective optimization algorithms.

2.2.1 Grasp Planning with Fixed End Effectors

Most robotic grasping research treats the end-effector geometry as a fixed input and searches for a feasible grasp pose on the workpiece. Zurbrügg et al. [72], for example, optimize grasp selection while assuming that the geometry and contact configuration of the gripper have already been defined. Analytical grasp-planning methods evaluate candidate contacts using geometric, force-closure, or form-closure conditions, whereas data-driven methods estimate grasp quality from training data. Despite their methodological differences, both approaches primarily vary the grasp pose rather than the gripper geometry.

Although Todescato et al. [5] establish gripper geometry as an optimization variable, their suction-gripper formulation is limited to a single suction cup. The design of a multi-suction-cup layout introduces additional decisions concerning the number and spatial arrangement of the cups, as well as a trade-off between grasp availability and the geometric distribution of valid contacts. This layout-optimization problem is addressed in this thesis.

The studies therefore address complementary but different questions. Grasp planning determines how an existing gripper should interact with a workpiece, whereas gripper-configuration optimization determines which geometry should be available before grasp planning is performed. Improving the grasp planner cannot fully compensate for a suction layout that cannot establish sufficient seals or for a jaw contour that cannot reach suitable contact regions. Consequently, fixed-gripper grasp-planning results do not directly resolve the end-effector design problem considered in this thesis.

2.2.2 Comparison of Multi-Suction Grasping Studies

Existing studies on multi-suction grippers address grasp prediction, experimental stability assessment, and load-aware motion planning, but generally assume predefined suction-cup layouts. Schillinger et al. [73] predict grasp quality from RGB-D images and optimize the grasp pose and gripper selection from a predefined set of multi-suction grippers. Their method selects among existing layouts rather than optimizing the positions of the individual cups.

Geng et al. [74] experimentally investigate how suction-cup size, geometric arrangement, centre of gravity offset, and robot acceleration influence grasp stability. Although different arrangements are evaluated, these are predefined experimental configurations rather than layouts generated through multi-objective optimization.

Lee et al. [75] model the load distribution among multiple suction cups using a quadratic program. They incorporate the resulting grasp-failure constraints into time-optimal trajectory planning to support the fast and reliable manipulation of heavy objects. Their work focuses on load prediction and motion planning for an existing gripper configuration, not on optimizing the cup arrangement.

These studies therefore address complementary aspects of multi-suction grasping. Schillinger et al. focus on grasp-pose and gripper selection, Geng et al. examine the effects of gripper configuration and handling conditions experimentally, and Lee et al. model load distribution during dynamic manipulation. However, none of these studies formulates the positions of the individual suction cups as decision variables in a multi-objective layout-optimization problem.

2.2.3 Comparison of Two-Finger Gripper Studies

Research on two-finger gripper design includes the optimization of compliant fingers, under-actuated mechanisms, and task-specific jaw geometries. Sun et al. [76] use three-dimensional topology optimization to design compliant continuum fingers and a lightweight gripper base. The resulting fingers deform passively to adapt to objects with different shapes. Similarly, Xie et al. [77] optimize the topology of flexible fingers while accounting for contact behaviour and contact-stress constraints. Their objective is to achieve adaptive, damage-free grasping of fragile and geometrically variable objects. Both studies focus on structural compliance and deformation, whereas this thesis considers the geometric optimization of rigid jaw contours.

Dong et al. [78] optimize the geometry and transmission system of an underactuated tendon-driven gripper. Their design variables include phalange dimensions, palm width, joint-mandrel radii, and tendon-routing parameters. The optimization aims to improve force transmission and grasp stability. Although this approach optimizes gripper geometry, it concerns the dimensions and actuation mechanism of an articulated gripper rather than the contact contour of a rigid parallel jaw.

This thesis addresses a different design problem by representing the contours of a rigid parallel-jaw gripper through a spline-based parameterization. The grasp points on the workpiece are predefined, and each candidate jaw contour is evaluated separately at every grasp point according to its conformity with the local workpiece geometry. Consequently, a workpiece with n predefined grasp points produces n conformity objectives and an n -dimensional objective space.

The optimization therefore does not select grasp points, introduce compliant deformation, or modify an articulated transmission mechanism. Instead, it seeks a rigid jaw contour that provides a suitable compromise in conformity across all predefined grasp points. The reviewed studies provide relevant approaches to automated gripper design, but they do not formulate rigid spline-based jaw-contour design as a multi-objective optimization problem in which each predefined grasp point contributes an individual objective.

2.2.4 Comparative Studies of Multi-Objective Optimization Algorithms

Comparative studies show that optimizer performance depends strongly on the problem and experimental setup. Tanabe and Ishibuchi [79] evaluate evolutionary algorithms on a suite of real-world multi-objective problems with varied characteristics, while Tian et al. [80] propose selecting algorithms based on problem features. Ishibuchi et al. [81] further demonstrate that termination conditions, population sizes, performance indicators, and test problems can change algorithm rankings. Audet et al. [20] emphasize that indicators assess different properties of Pareto-front approximations, such as convergence, distribution, and spread.

For expensive optimization, Qin et al. [82] compare two combinations of evolutionary and Bayesian optimization on ZDT and DTLZ problems and find that Bayesian evolutionary optimization is more efficient under their experimental conditions. Santoni et al. [12] compare high-dimensional Bayesian optimization methods on single-objective BBOB functions and report that performance varies with dimensionality, function landscape, and evaluation budget. These findings are therefore specific to the investigated problems and settings.

Such results cannot be transferred directly to the collision-aware suction-layout and spline-based jaw-contour problems considered in this thesis. Therefore, this thesis compares selected multi-objective evolutionary and Bayesian algorithms under common evaluation budgets, initialization conditions, repeated runs, and statistical procedures. The aim of this comparison is to identify methods suitable for these specific gripper-design problems rather than a universally superior optimizer.

2.2.5 Comparative Synthesis and Research Gap

The reviewed literature reveals three related research gaps. First, grasp-planning studies generally optimize grasp poses for predefined end-effectors rather than the gripper configuration itself. Second, existing multi-suction studies assume predefined cup layouts, while the reviewed two-finger studies focus on compliant structures or underactuated mechanisms rather than rigid spline-based jaw contours. They therefore do not address the suction-layout and jaw-contour optimization problems considered in this thesis. Third, existing algorithm comparisons do not determine which optimization methods are best suited to these collision-aware geometric design problems.

For the multi-suction gripper, the unresolved problem is to balance the availability of feasible contacts with their geometric distribution by optimizing the positions of the individual cups. For the two-finger gripper, it is to design a rigid jaw contour that provides suitable conformity across predefined grasp points while satisfying geometric and collision constraints. Both are expensive multi-objective black-box problems, but their different representations and evaluation procedures may favour different optimization methods.

Within the literature reviewed for this thesis, no controlled comparison of evolutionary and Bayesian multi-objective algorithms was identified for these two gripper-design formulations.

2.2.6 Research Positioning and Contribution

This thesis addresses these gaps by treating gripper geometry as the primary optimization variable. The multi-suction formulation optimizes the spatial arrangement of individual suction cups, whereas the two-finger formulation optimizes spline-based contours for rigid parallel jaws. Candidate designs are assessed using problem-specific geometric and collision-aware evaluations.

A modular framework separates geometric evaluation from optimization through a common interface. This allows the applicable algorithms to operate on the same problem formulation, constraints, and evaluation conditions. Selected evolutionary and Bayesian multi-objective algorithms are compared under defined evaluation budgets using repeated independent runs, convergence histories, multi-objective performance indicators, and statistical tests. Evaluation counts and optimizer overhead are reported separately where computational cost is relevant.

The contribution of this thesis is therefore a problem-specific comparison rather than a claim of universal algorithm superiority. The results provide evidence for selecting optimization methods for the suction-layout and rigid-jaw-contour formulations considered here.

3 Methodology

This chapter describes the methodological decisions, theoretical frameworks, and evaluation strategies underlying the gripper configuration optimization study. It covers the problem formulation, the choice of optimization library and algorithms, the evaluation architecture, the workpiece selection rationale, and the comparative experimental design. Implementation details and experimental procedures are treated separately in the approach chapter.

3.1 Research Design

3.1.1 Research Paradigm

The thesis adopts a computational experimental methodology. Both gripper configuration problems are approached through mathematical modeling, computational-based evaluation, and controlled comparative optimization experiments. This design is appropriate because the objectives governing gripper quality cannot be expressed in closed analytical form; they depend on geometric interactions between the gripper and the workpiece that must be assessed computationally. The core methodological contribution is not the development of new optimization algorithms, but the rigorous comparative evaluation of existing algorithms on well-defined, practically motivated black-box problems.

3.1.2 Study Structure

The two gripper types are treated as independent optimization studies. The suction-cup gripper optimization seeks a cup layout that simultaneously maximizes grasp stability and grasp set diversity. The two-finger gripper optimization seeks a jaw contour that achieves high contact quality across a set of representative workpieces; since the evaluation pipeline

computes a separate proximity-based score for each workpiece, each workpiece contributes one objective, making this a multi-objective problem whose dimensionality scales with the number of workpieces selected. In both cases, the problem structures and objective spaces differ, and the algorithm comparison is therefore conducted separately for each gripper type with results not aggregated across them.

3.1.3 Experimental Setup

For each problem, a fixed set of candidate algorithms is applied to a set of representative workpieces. Each algorithm is executed once per workpiece under identical conditions, including a shared evaluation budget and consistent initialization settings[83]. Algorithm performance is then analyzed and interpreted across all workpieces and both gripper types, with particular focus on three aspects: convergence behaviour over the evaluation budget, Pareto front quality as measured by established performance indicators, and the sensitivity of algorithm rankings to workpiece geometry and objective space dimensionality[20]. This design allows conclusions to be drawn about which algorithms are best suited to each problem class and how performance varies with problem instance characteristics[84].

3.1.4 Scope and Limitations

Since no repeated independent runs are performed, the study does not support statistical inference about algorithm performance distributions. Conclusions are therefore descriptive and comparative in nature, grounded in the observed results across the workpiece set rather than in probabilistic significance testing. Furthermore, the evaluation relies entirely on computational calculation; physical validation of the optimized gripper configurations is outside the scope of this work.

3.2 Problem Formulation

3.2.1 Multi-Objective Optimization Framework

Both gripper configuration problems are formulated as multi-objective optimization problems. Let $\mathbf{q} \in \mathcal{Q}$ denote the decision vector and let d denote the number of decision variables.

The general problem is formulated as

$$\min_{\mathbf{q} \in \mathcal{Q}} \mathbf{f}(\mathbf{q}) = (f_1(\mathbf{q}), f_2(\mathbf{q}), \dots, f_m(\mathbf{q})), \quad (3.1)$$

where m is the number of objectives, subject to inequality constraints

$$g_j(\mathbf{q}) \leq 0, \quad j = 1, \dots, n_c, \quad (3.2)$$

and variable bounds

$$q_i^L \leq q_i \leq q_i^U, \quad i = 1, \dots, d. \quad (3.3)$$

Objectives that are naturally maximized are negated to maintain a consistent minimization framework throughout. A multi-objective formulation is preferred over a weighted scalar aggregate because it produces a Pareto-optimal solution set that makes trade-offs explicit and preserves design alternatives without requiring the prior specification of preference weights.

The number of objectives m is directly determined by the number of workpieces used in the evaluation. This design choice has two methodological motivations. First, it forces the optimizer to find gripper configurations that perform well across geometrically diverse workpieces simultaneously, rather than overfitting to a single object. Second, it allows the study to examine algorithm behaviour as the dimensionality of the objective space increases, since adding workpieces directly increases m .

3.2.2 Parameterization of Gripper Geometry

A fundamental methodological decision is how to represent gripper geometry as a finite-dimensional decision vector suitable for black-box optimization [85] [5]. For both gripper types, the representation must satisfy three competing demands: it must be expressive enough to capture meaningful geometric variation, compact enough to keep the search space tractable, and physically interpretable so that every point in the search space corresponds to a realizable gripper configuration [86].

For the suction-cup gripper, the most general representation would specify the full Cartesian coordinates of each cup independently, yielding a six-dimensional search space for three cups in the plane. However, this representation introduces redundancy through rigid-body

symmetries and admits degenerate configurations such as collinear or overlapping cups. An alternative is to exploit the geometric structure of the problem: the primary geometric quantity governing stability is the area of the triangle formed by the three cups, which depends on the radial spread and the relative angular spacing of the cups rather than their absolute positions. This motivates a reduced polar parameterization that encodes the common radial distance and the three angular positions, resulting in a four-dimensional search space that eliminates translational and overall rotational redundancy while retaining the degrees of freedom that directly affect the stability and collision behavior of the layout.

For the two-finger gripper, the finger jaw contour must be represented in a way that allows smooth, physically realizable shapes to be generated from a small number of parameters [87]. A spline parameterization is adopted because it provides a compact, smooth, and easily constrained contour representation, with fixed boundary control points enforcing the geometric constraints at the finger tip and base while the remaining interior control points form the free decision variables. The number of free control points can therefore be chosen directly to balance geometric expressiveness against search space dimensionality.

3.2.3 Selection of Objective Functions and Evaluation Metrics

The choice of objective functions determines what constitutes a good gripper configuration and therefore directly shapes the optimization problem [5, 87]. For each gripper type, the objectives must reflect physically meaningful performance criteria while remaining computable from geometry alone, since no physical simulation or hardware trial is available during optimization.

For the suction-cup gripper, two competing criteria govern layout quality. Grasp stability under external disturbances is classically characterized by force-closure metrics or grasp wrench space volume, both of which require contact force modeling and friction assumptions[88]. Since the optimization operates on geometry alone and no contact force model is available during the search, the area of the support triangle formed by the three cup positions is used as the stability objective. A larger triangle area directly increases the moment arm resisting tipping disturbances, making it a geometrically motivated and parameter-free indicator of layout quality[89]. The second criterion is the availability of valid grasps on the target workpiece[90]. A simple count of collision-free grasp poses does not capture the practical

utility of the grasp set, since many similar poses offer little additional coverage. Diversity-based metrics, which measure the spread of valid grasps in both position and orientation, are better indicators of operational flexibility and have been used in related work on grasp set evaluation[91]. A weighted combination of spatial and orientational diversity is therefore adopted, with both components normalized to enable consistent comparison across workpieces of different scales.

For the two-finger gripper, the design objective is to find a jaw contour that achieves close contact with the workpiece across as many placements as possible without collision[85]. The evaluation is performed using a voxel-based convolution scoring function developed at Fraunhofer IPA, which evaluates finger-to-workpiece proximity over all candidate placements without requiring explicit grasp candidate enumeration, and excludes colliding placements by construction[92]. This function produces a single scalar score per workpiece that directly serves as the optimization objective.

3.3 Optimization Library Selection

The evolutionary multi-objective optimization algorithms are implemented and executed using the `pymoo` framework. This choice is motivated by several methodological considerations. `pymoo` is a widely used Python framework for multi-objective evolutionary optimization and provides a unified implementation of numerous algorithms. This reduces implementation bias and helps ensure that observed performance differences reflect genuine algorithmic properties rather than differences in code quality or implementation choices. It also includes standardized performance indicators, such as hypervolume, inverted generational distance (IGD), and generational distance (GD), as well as built-in constraint-handling and visualization tools. These features improve the reproducibility and comparability of the results. Furthermore, `pymoo` is actively maintained, open-source, and thoroughly documented, making it easier for other researchers to replicate and extend the evolutionary optimization experiments conducted in this thesis [18].

Bayesian multi-objective optimization is implemented using the `Ax` platform, with its underlying modeling and acquisition-function capabilities provided by `BoTorch`. `Ax` provides a high-level interface for defining parameter spaces, configuring multiple objectives, managing trials, and extracting the resulting Pareto frontier. Its integration with `BoTorch` enables model-based optimization using probabilistic surrogate models, typically Gaussian processes,

together with acquisition functions designed for multi-objective optimization. In the implementation used in this work, **Ax** automatically applies an initialization strategy based on the center of the search space and Sobol-generated samples before transitioning to model-based Bayesian optimization. This approach provides initial coverage of the search space before the surrogate model is used to select subsequent candidates [3].

The use of **Ax** and **BoTorch** avoids the need to implement surrogate-model fitting, uncertainty estimation, and acquisition-function optimization manually. It therefore reduces implementation bias and provides a robust and reproducible Bayesian optimization workflow. Both libraries are open-source, actively maintained, and designed to support extensible experimentation with modern Bayesian optimization methods [3, 19]. Consequently, **pymoo** is used for the evolutionary algorithms, whereas **Ax** and **BoTorch** are used for Bayesian multi-objective optimization.

3.4 Algorithm Selection

The selection of optimization algorithms from the set available in **pymoo** was guided by two criteria: suitability for the problem class and coverage of fundamentally different optimization paradigms.

Regarding suitability, the gripper configuration problems are continuous, constrained, black-box multi-objective optimization problems. This rules out a substantial portion of the algorithms available in **pymoo**. Many algorithms in the library are designed exclusively for single-objective optimization, including genetic algorithms, differential evolution, particle swarm optimization, CMA-ES, evolution strategies, and direct search methods, and are therefore not applicable. Several multi-objective algorithms are designed for specialized problem settings outside the scope of this thesis. These include algorithms for dynamic optimization, where objectives or constraints change over time, and preference-based algorithms, which require user-defined preference information to guide the search toward specific regions of the Pareto front. Since both gripper problems are static and the goal is to approximate the full Pareto front without incorporating decision-maker preferences, these specialized methods were excluded[18].

Among the remaining general-purpose multi-objective algorithms, the selected methods were chosen because they represent fundamentally different optimization strategies rather than incremental variations of the same concept:

- **NSGA-II** represents Pareto-dominance-based selection and serves as the standard baseline for multi-objective evolutionary optimization[67].
- **NSGA-III** extends the dominance-based approach through reference directions, making it better suited to problems with a larger number of objectives, as arises when more workpieces are included[68].
- **MOEA/D** represents decomposition-based optimization, solving a set of scalar sub-problems rather than relying solely on Pareto dominance[69].
- **SMS-EMOA** represents indicator-based optimization, directly maximizing the hypervolume indicator and thereby emphasizing Pareto front quality[70].
- **CMOPSO** provides a swarm-intelligence-based approach, employing a competitive particle swarm mechanism to balance exploration and convergence[71].

Together, these five algorithms cover the major paradigms of general-purpose multi-objective optimization, enabling a comparison of fundamentally different search strategies on the same gripper configuration problems[93, 94, 95].

In addition to the evolutionary algorithms, multi-objective Bayesian optimization is included as a comparative strategy. Bayesian optimization is motivated by its sample efficiency in expensive black-box settings: a probabilistic surrogate model is maintained over the objective functions and an acquisition function is used to guide evaluations toward promising regions of the search space. Its inclusion allows the study to assess whether surrogate-based methods offer advantages over population-based methods under the evaluation budgets considered.

3.5 Workpiece Selection

The workpieces were selected to provide geometric diversity rather than to optimize performance for a particular object. Since the objective of this thesis is to compare optimization

algorithms, it is essential that the benchmark problems exhibit different geometric characteristics and consequently different optimization landscapes[96, 97].

The **IPAIntermediatePlate** represents a simple industrial workpiece with large planar surfaces and regular geometry, reflecting many components commonly encountered in manufacturing and material handling. Such geometry typically provides a relatively large feasible grasping region and a smoother objective landscape. In contrast, the **Stanford Bunny** exhibits a highly irregular three-dimensional shape with varying surface curvature and complex local features, resulting in a more challenging optimization problem characterized by narrower feasible regions, stronger collision constraints, and a potentially more multimodal objective landscape[98].

By evaluating all algorithms on both workpieces simultaneously, the study reduces the risk of drawing conclusions that are specific to a single geometry and enables a more robust assessment of algorithm performance across problems of varying complexity. The use of multiple workpieces as separate objectives also means that the comparison implicitly assesses how well each algorithm handles increasing objective space dimensionality[96].

3.6 Evaluation Architecture

The evaluation of candidate gripper configurations is decoupled from the optimization process through a client-server architecture. This separation constitutes a deliberate methodological design choice that enforces a strict black-box interface between the optimization strategy and the objective function computation. The optimization algorithms operate solely on input-output pairs $(\mathbf{q}, \mathbf{f}(\mathbf{q}))$ without access to internal evaluation logic[99]. This architecture also ensures that any optimization algorithm conforming to the interface can be applied to either evaluation pipeline without modification of either component, supporting reproducibility and extensibility.

3.7 Comparative Experimental Design

A controlled comparison protocol is adopted to ensure that observed performance differences are attributable to the optimization methods rather than to differences in experimental

conditions. All algorithms are evaluated on the same workpieces, gripper parameterizations, objective functions, feasibility conditions, and evaluation budget. Each algorithm is executed under multiple independent random seeds to assess not only mean performance but also variability and robustness across runs[100]. Algorithm hyperparameters are initialized from the default values provided by the `pymoo` framework [18]. Where the optimization pipeline is configured to do so, a structured hyperparameter tuning procedure is executed prior to the main optimization run, replacing the defaults with problem-specific settings and reducing the risk of performance bias from incidental parameter choices. [101].

3.8 Performance Indicators

Algorithm performance is assessed using standard multi-objective performance indicators. The primary indicator is the *hypervolume indicator*, which measures the volume of objective space dominated by the solution set with respect to a reference point, thereby jointly capturing convergence quality and Pareto front diversity[20]. Convergence profiles over function evaluations are also recorded to assess the efficiency of each algorithm within the given evaluation budget. Robustness is characterized by the distribution of indicator values across repeated independent runs[102].

4 Implementation

4.1 Implementation of Multi-Suction Cup Gripper Optimization

This chapter presents the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The framework supports both evolutionary multi-objective optimization and Bayesian multi-objective optimization. The evolutionary algorithms are implemented using the `pymoo` library [18], whereas Bayesian optimization is implemented using the `Ax` platform and its underlying `BoTorch` modeling framework [3, 19].

For both optimization approaches, the evaluation of candidate gripper layouts is delegated to an external service through an API. This separation provides a modular architecture in which the optimization strategy and the domain-specific grasp evaluation remain decoupled. Consequently, the same decision-variable representation, grasp-generation procedure, and objective functions can be used by both evolutionary and Bayesian optimization methods, allowing their performance to be compared under a common evaluation pipeline.

4.1.1 Reduced Parameterization of Multi-Suction Cup Gripper

The gripper consists of three suction cups whose positions must ultimately be represented in Cartesian space in order to perform grasp generation, collision checking, and objective computation. However, the Cartesian suction cup coordinates are not used directly as optimization variables. Instead, the layout is parameterized in a reduced form to constrain the search space and encode geometric structure.

All three suction cups are constrained to lie on a common circle in the gripper plane. Their configuration is therefore defined by one shared radius r and three angular parameters

$\alpha_1, \alpha_2, \alpha_3$:

$$\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3] \quad (4.1)$$

The angular parameters determine the positions of the suction cups on the circle, while the common radius determines their distance from the gripper center. For a fixed gripper height z_0 , the Cartesian coordinates of the suction cups are obtained through the polar-to-Cartesian transformation

$$\mathbf{c}_i = \begin{bmatrix} r \cos(\alpha_i) \\ r \sin(\alpha_i) \\ z_0 \end{bmatrix}, \quad i = 1, 2, 3. \quad (4.2)$$

The resulting Cartesian gripper configuration is written as

$$\mathbf{p} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3]. \quad (4.3)$$

Thus, the optimization is carried out in the reduced 4-dimensional parameter space $\mathbf{q}$, while the grasp evaluation operates on the derived Cartesian coordinates $\mathbf{p}$.

This formulation reduces the dimensionality of the search space from six planar degrees of freedom,

$$[(x_1, y_1), (x_2, y_2), (x_3, y_3)] \in \mathbb{R}^6, \quad (4.4)$$

to four variables,

$$[r, \alpha_1, \alpha_2, \alpha_3]. \quad (4.5)$$

The reduction improves search efficiency and excludes many geometrically redundant or undesirable layouts while preserving sufficient flexibility for diverse triangular suction cup arrangements.

Assumption A triangular configuration of three suction cups is assumed to yield a stable grasp [89]. This is motivated by the fact that three spatially distributed contact points form a statically stable support polygon, enabling effective load distribution and resistance to external disturbances.

4.1.2 Decision Variables and Bounds for Multi-Suction Cup Gripper

The optimization is performed over the reduced decision vector

$$\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3]. \quad (4.6)$$

All variables are treated as continuous. The angular variables are bounded by

$$\alpha_i \in [0, 2\pi), \quad i = 1, 2, 3, \quad (4.7)$$

and the common radius is bounded by

$$r \in [r_{\min}, r_{\max}]. \quad (4.8)$$

The upper bound $r_{\max}$ is determined from the bounding box of the largest workpiece considered during optimization, ensuring that the gripper footprint remains compatible with the object dimensions. The lower bound $r_{\min}$ is determined from the diameter of an individual suction cup, ensuring that the configuration remains physically realizable and that the suction cups do not collapse into an excessively compact arrangement.

All optimization methods generate candidate values within these bounds. In the evolutionary algorithms, the initial population is obtained by random sampling and subsequently modified through the operators of the selected algorithm. In Bayesian optimization, Ax first evaluates the center of the search space and then uses Sobol sampling to generate an initial space-filling design. After sufficient initialization data have been collected, Ax transitions to model-based Bayesian optimization.

4.1.3 Optimization Framework of Multi-Suction Cup Gripper

The multi-suction cup gripper problem is solved using either an evolutionary multi-objective optimization algorithm or Bayesian multi-objective optimization. Both approaches operate on the same reduced decision vector, variable bounds, objective functions, and external evaluation service.

The common optimization configuration contains:

- Number of objectives n_{obj}
- Decision-variable definitions and bounds
- Optimization direction
- Choice of optimization method
- Evaluation endpoint URL

For evolutionary optimization, the additional configuration parameters include:

- Population size N
- Maximum number of generations G_{max}
- Choice of evolutionary algorithm
- Algorithm-specific parameters, such as crossover and mutation settings

For Bayesian optimization, the additional configuration parameters include:

- Number of optimization batches B_{max}
- Maximum number of candidates per batch b
- Bayesian generation strategy

In the current Bayesian implementation, the configuration value named **generations** represents the number of candidate-generation batches rather than evolutionary generations. The maximum requested number of Bayesian evaluations is therefore

$$E_{\max}^{\text{BO}} = B_{\max} b. \quad (4.9)$$

The actual number may be smaller if Ax returns fewer than b candidates in a particular batch.

Supported Optimization Methods

The framework supports the following evolutionary and Bayesian multi-objective optimization methods:

Table 4.1: Supported multi-objective optimization methods.

Family	Identifier	Method
Evolutionary	NSGA2	Non-dominated Sorting Genetic Algorithm II [67]
Evolutionary	NSGA3	Non-dominated Sorting Genetic Algorithm III [68]
Evolutionary	MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [69]
Evolutionary	CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
Evolutionary	SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [70]
Bayesian	AX-MBM	Ax modular BoTorch model for multi-objective Bayesian optimization [3, 19]

For evolutionary algorithms requiring reference directions, such as NSGA-III and MOEA/D, uniform reference directions are generated using the method of Das and Dennis [52]:

$$\Lambda = \text{get_reference_directions}(\text{"das-dennis"}, n_{\text{obj}}, n_{\text{partitions}}). \quad (4.10)$$

The Bayesian implementation uses the default Ax generation strategy identified at runtime as **Center+Sobol+MBM:fast**. This strategy consists of three stages:

1. Evaluation of the center of the bounded search space

2. Sobol quasi-random sampling for initialization
3. Model-based optimization using the Ax modular BoTorch model (MBM)

The Center and Sobol stages do not employ a surrogate model or acquisition function. After transitioning to the MBM stage, Ax fits probabilistic surrogate models to the completed observations and selects new candidates using an acquisition function appropriate for the multi-objective problem. For continuous multi-objective problems, this generally involves Gaussian-process surrogates and a hypervolume-improvement acquisition function. The exact model and acquisition-function classes are selected by Ax according to the installed version and experiment configuration and should therefore be recorded from the runtime configuration to ensure reproducibility.

4.1.4 Initialization of Multi-Suction Cup Gripper Optimization

The initialization procedure depends on the selected optimization approach.

Evolutionary Initialization

For evolutionary optimization, an initial population of N candidate solutions is generated by sampling the reduced decision variables within their bounds:

$$\mathcal{P}^{(0)} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N\}. \quad (4.11)$$

Each candidate decision vector $\mathbf{q}_k$ is transformed into its corresponding Cartesian gripper configuration $\mathbf{p}_k$ before objective evaluation.

Bayesian Initialization

Bayesian optimization does not maintain an evolutionary population. Instead, it stores a dataset of completed trials:

$$\mathcal{D}_t = \{(\mathbf{q}_k, \mathbf{g}(\mathbf{q}_k))\}_{k=1}^t, \quad (4.12)$$

where $\mathbf{q}_k$ is an evaluated gripper parameterization and $\mathbf{g}(\mathbf{q}_k)$ is its observed objective vector.

The Ax generation strategy first evaluates the center of the search space and then generates an initial set of candidates using Sobol quasi-random sampling. Sobol sampling provides broad coverage of the bounded decision space before a surrogate model is fitted. Once the initialization requirements of the generation strategy have been satisfied, Ax transitions to model-based candidate generation.

4.1.5 Objective Function Evaluation of Multi-Suction Cup Gripper via API

The evaluation of objective values is decoupled from the optimization loop and delegated to an external evaluation service. For each candidate solution $\mathbf{q}$, the corresponding Cartesian suction cup coordinates are first computed by transforming the radial and angular parameters into 3D positions. The resulting Cartesian gripper configuration is then passed to the evaluation endpoint, which executes the grasp evaluation procedure and returns the objective values.

This process can be written as

$$\mathbf{p} = T(\mathbf{q}), \quad \mathbf{f}(\mathbf{q}) = \mathbf{f}(T(\mathbf{q})) = \text{API_call}(\mathbf{p}) \rightarrow [f_1(\mathbf{p}), f_2(\mathbf{p})]. \quad (4.13)$$

where $T(\cdot)$ denotes the transformation from the reduced parameterization to Cartesian suction cup coordinates.

In the implementation, this transformation is carried out programmatically in the evaluation pipeline before grasp generation and objective computation. For a candidate radius and list of angles, the suction cup coordinates are constructed as

$$\mathbf{c}_i = (r \cos(\alpha_i), r \sin(\alpha_i), z_0), \quad i = 1, 2, 3. \quad (4.14)$$

This architecture enables:

- Separation of concerns between optimization logic and domain-specific evaluation
- Scalability through independent deployment of the evaluation service

- Reusability of the optimization framework for different problem domains

Grasp Evaluation

The evaluation service receives a Cartesian gripper configuration $\mathbf{p} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3]$ and computes the associated grasp quality measures through the following procedure.

1. Candidate Grasp Generation and Feasibility Filtering

Suction grasp candidates are generated using a method developed by Fraunhofer IPA, following the automatic suction grasp generation procedure described by [37]. Points are sampled on the surface of the workpiece, and each sampled point defines a candidate grasp through its position and local surface normal. The local surface normal determines the approach direction of the suction cup.

In the original method, each suction grasp candidate is tested for feasible seal formation using a quasi-static spring model of a suction cup, as introduced in [103]. If the grasp point is considered feasible, the candidate is further tested with a gripper collider model for collisions between the gripper and the object.

In this thesis, the method was extended to a gripper with three suction cups. For each candidate gripper pose, the seal-formation test is performed for all three suction cup contact positions $\mathbf{c}_1$, $\mathbf{c}_2$, and $\mathbf{c}_3$. A candidate is considered feasible only if all three suction cups form a sufficient seal. The candidate is then checked for collisions in simulation, and only collision-free candidates are retained as valid grasps.

2. Clustering of Grasps

The set $\mathcal{G}_{\text{valid}}$ often contains many grasps that are nearly identical in position and orientation. When the number of valid grasps $m = |\mathcal{G}_{\text{valid}}|$ exceeds a threshold `nb_grasps_clustering`, a farthest-point clustering procedure is applied to select a diverse reduced subset. Otherwise, all valid grasps are used directly.

Each grasp g_i is represented by a pose descriptor

$$\mathbf{z}_i = [\text{vec}(\mathbf{R}_i)^\top, \mathbf{t}_i^\top]^\top \in \mathbb{R}^{12}, \quad (4.15)$$

where $\mathbf{R}_i \in SO(3)$ is the orientation and $\mathbf{t}_i \in \mathbb{R}^3$ is the position. Two distances are used: the translational distance

$$d_{\text{pos}}(i, j) = \|\mathbf{t}_i - \mathbf{t}_j\|_2 \quad (4.16)$$

and the orientation distance

$$d_{\text{ori}}(i, j) = \frac{1}{9} \|\text{vec}(\mathbf{R}_i) - \text{vec}(\mathbf{R}_j)\|_2^2. \quad (4.17)$$

Cluster centers are selected greedily by farthest-point sampling in position space. The first center is the grasp with the largest mean translational distance to all others, and each subsequent center maximizes its minimum distance to the already selected centers. This promotes broad spatial coverage.

Optionally, orientation diversity is added within each center's local neighborhood. For a selected center m_k , grasps within a spatial radius δ_{intra} form a neighborhood, from which up to q_{intra} additional grasps are chosen by farthest-point sampling on d_{ori} , capturing distinct orientations at similar positions.

The resulting clustered subset

$$\mathcal{G}_{\text{clustered}} \subseteq \mathcal{G}_{\text{valid}} \quad (4.18)$$

is then used for diversity evaluation.

3. Triangle Area Calculation

The area of the triangle formed by the three suction cups is computed from their Cartesian coordinates. If the planar coordinates of the three cups are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , then

$$A(\mathbf{p}) = \frac{1}{2} |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)|. \quad (4.19)$$

This objective favors gripper layouts with a wider support polygon, which is associated with improved grasp stability.

4. Diversity Metric Calculation

A diversity score $D(\mathbf{p})$ is computed to quantify how broadly the resulting valid grasp set is distributed in position and orientation.

Let the set used for diversity evaluation contain n grasps. If clustering is performed, then

$$n = |\mathcal{G}_{\text{clustered}}|, \quad (4.20)$$

otherwise

$$n = |\mathcal{G}_{\text{valid}}|. \quad (4.21)$$

Let the corresponding grasp positions be

$$\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \quad (4.22)$$

and the corresponding orientation matrices be

$$\{\mathbf{R}_1, \dots, \mathbf{R}_n\}. \quad (4.23)$$

To normalize the spatial term with respect to object size, a characteristic object scale is defined as the Euclidean norm of the axis-aligned bounding-box extents:

$$s = \|\mathbf{e}\|_2, \quad (4.24)$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the bounding-box side lengths.

The spatial component of diversity is computed as the mean pairwise Euclidean distance between grasp positions:

$$D_{\text{pos}}(\mathbf{p}) = \frac{1}{s} \cdot \frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^n \|\mathbf{y}_i - \mathbf{y}_j\|_2. \quad (4.25)$$

The orientation component is computed from the mean pairwise geodesic distance between rotation matrices. For two grasp orientations $\mathbf{R}_a, \mathbf{R}_b \in SO(3)$, the relative

rotation is

$$\Delta \mathbf{R} = \mathbf{R}_a^\top \mathbf{R}_b \quad (4.26)$$

and the associated geodesic distance is

$$d_{\text{geo}}(\mathbf{R}_a, \mathbf{R}_b) = \cos^{-1} \left(\frac{\text{tr}(\Delta \mathbf{R}) - 1}{2} \right). \quad (4.27)$$

For numerical stability, the argument of $\cos^{-1}$ is clipped to the interval $[-1, 1]$.

The normalized orientation diversity is then defined as

$$D_{\text{rot}}(\mathbf{p}) = \frac{1}{\pi} \cdot \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} d_{\text{geo}}(\mathbf{R}_i, \mathbf{R}_j). \quad (4.28)$$

The final diversity score is computed as the equally weighted combination

$$D(\mathbf{p}) = \text{clip} (0.5 D_{\text{pos}}(\mathbf{p}) + 0.5 D_{\text{rot}}(\mathbf{p}), 0, 1). \quad (4.29)$$

Continuity considerations Clustering is only applied when the number of valid grasps exceeds the predefined threshold `nb_grasps_clustering`. If fewer valid grasps are available, clustering is skipped and the diversity metric is evaluated directly on the valid grasp set. Therefore, the omission of clustering does not by itself introduce a discontinuity in the objective function.

A discontinuity may arise only in the degenerate case where fewer than two valid grasps remain after collision filtering. Since both the spatial and orientation components of the diversity metric are based on pairwise distances, these terms cannot be evaluated meaningfully for $n < 2$. In that case, the corresponding terms are set to zero. As a result, small changes in the gripper configuration may cause an abrupt transition between a nonzero diversity score and a degenerate zero-valued diversity contribution.

To reduce the likelihood of this case, the number of initially sampled surface points was increased from 100 to 1000. The denser surface sampling produces a larger set of

candidate grasps and makes it much less likely that fewer than two valid grasps remain after filtering. Consequently, the diversity metric is evaluated on a sufficiently rich grasp set in the vast majority of cases, yielding a more stable optimization objective in practice.

Objective Evaluation

After grasp evaluation, each candidate layout is assigned the natural objective vector

$$\mathbf{g}(\mathbf{q}) = [A(\mathbf{p}), D(\mathbf{p})], \quad \mathbf{p} = T(\mathbf{q}). \quad (4.30)$$

Both components are maximized. A larger triangle area is associated with a wider support polygon and improved grasp stability, while a larger diversity score indicates that the valid grasps are more broadly distributed in position and orientation.

The objective-direction convention is handled differently by the two optimization frameworks. Since `pymoo` assumes minimization, the evolutionary algorithms negates objective vector

$$\mathbf{f}_{\text{EA}}(\mathbf{q}) = [-A(\mathbf{p}), -D(\mathbf{p})]. \quad (4.31)$$

In the Bayesian implementation, the external evaluation service values are left as is, since `Ax` assumes maximization

$$\mathbf{f}_{\text{BO}}(\mathbf{q}) = [A(\mathbf{p}), D(\mathbf{p})]. \quad (4.32)$$

The optimization direction is specified through the `Ax` objective expression. Consequently, the observations themselves are not negated before being reported to `Ax`. This preserves the original physical meaning of the objective values in the returned Pareto set.

4.1.6 Optimization Loop for Multi-Suction Cup Gripper

The candidate-generation process differs between evolutionary and Bayesian optimization, while the external candidate-evaluation procedure remains identical.

Evolutionary Optimization Loop

The evolutionary optimization proceeds according to

$$\mathcal{P}^{(t+1)} = \mathcal{A}(\mathcal{P}^{(t)}, \mathbf{f}_{\text{EA}}), \quad t = 0, 1, \dots, \quad (4.33)$$

where $\mathcal{A}$ denotes the selected evolutionary algorithm.

Depending on the selected algorithm, the new population is generated through operations such as selection, crossover, mutation, decomposition, particle movement, or survival based on hypervolume contribution. Every candidate in the resulting population is evaluated through the common API-based grasp evaluation pipeline.

Bayesian Optimization Loop

The Bayesian optimization procedure maintains the observation dataset $\mathcal{D}_t$. During the initialization phase, candidates are generated from the center of the search space and by Sobol sampling. During the model-based phase, the available observations are used to fit probabilistic surrogate models:

$$\hat{\mathbf{g}}_t = \mathcal{M}(\mathcal{D}_t), \quad (4.34)$$

where $\mathcal{M}$ denotes the surrogate-modeling procedure used by Ax and BoTorch.

The surrogate models provide predictions of the objective values together with their uncertainty. An acquisition function $\alpha(\mathbf{q} \mid \mathcal{D}_t)$ uses this information to balance the exploration of uncertain regions and the exploitation of regions expected to improve the current Pareto frontier.

For a requested batch size b , the next candidate batch is generated as

$$\mathcal{Q}_{t+1} = \arg \max_{\substack{\mathcal{Q} \subset \mathcal{X} \\ |\mathcal{Q}| \leq b}} \alpha(\mathcal{Q} \mid \mathcal{D}_t), \quad (4.35)$$

where $\mathcal{X}$ is the bounded decision space.

Each candidate $\mathbf{q} \in \mathcal{Q}_{t+1}$ is evaluated through the external evaluation service. The resulting observation is then supplied to $\mathbf{Ax}$ as a completed trial:

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{q}_{t+1}, \mathbf{g}(\mathbf{q}_{t+1}))\}. \quad (4.36)$$

The surrogate model is subsequently updated using the expanded dataset, and the process is repeated until the specified evaluation budget is exhausted. Although multiple candidates can be requested in one batch, the current implementation submits the candidates in that batch sequentially to the evaluation service.

4.1.7 Termination Criteria for Multi-Suction Cup Gripper Optimization

The termination criteria depend on the selected optimization approach.

Table 4.2: Termination criteria for evolutionary and Bayesian optimization.

Approach	Criterion	Symbol	Description
Evolutionary	Design-space tolerance	x_{tol}	Change in decision variables below threshold
Evolutionary	Objective-space tolerance	f_{tol}	Relative objective improvement below threshold
Evolutionary	Maximum generations	G_{max}	Maximum number of generations reached
Evolutionary	Maximum evaluations	E_{max}	Maximum number of function evaluations reached
Evolutionary	Hypervolume	HV_{max}	Hypervolume progression stagnates
Bayesian	Maximum batches	B_{max}	Maximum number of candidate-generation batches reached
Bayesian	Maximum evaluations	$E_{\text{max}}^{\text{BO}}$	Bayesian evaluation budget exhausted

For evolutionary optimization, the design-space and objective-space tolerances are evaluated over a sliding window of w generations.

In the current Bayesian implementation, termination is budget-based. The optimization performs at most $B_{\max}$ candidate-generation batches, with up to b candidates requested per batch. Therefore,

$$E_{\max}^{\text{BO}} = B_{\max} b. \quad (4.37)$$

No design-space or objective-space convergence tolerance is currently applied to the Bayesian optimization loop.

4.1.8 Performance Monitoring for Multi-Suction Cup Gripper

The hypervolume indicator [104] is tracked at each generation via a callback mechanism:

$$\text{HV}^{(t)} = \text{HV}(\mathcal{F}^{(t)}, \mathbf{r}) \quad (4.38)$$

Here, $\mathbf{r} \in \mathbb{R}^{n_{\text{obj}}}$ is the reference point and $\mathcal{F}^{(t)}$ denotes the set of objective vectors at generation t . A monotonically increasing hypervolume indicates convergence toward the Pareto front.

4.1.9 Optional Hyperparameter Optimization for Multi-Suction Cup Gripper Problem

The Evolutionary Optimization framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The mean hypervolume across runs serves as the meta-objective:

$$\min_{\theta} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \text{HV}(\mathcal{F}_s, \mathbf{r}) \right\} \quad (4.39)$$

The best hyperparameters are then applied to the main optimization run.

4.1.10 Output of Multi-Suction Cup Gripper Optimization

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{q} \mid \nexists \mathbf{q}' \in \mathcal{P} : \mathbf{f}(\mathbf{q}') \preceq \mathbf{f}(\mathbf{q})\}. \quad (4.40)$$

The final output includes:

- The Pareto-optimal set $\mathcal{P}^*$ in the reduced decision space, with corresponding objective values
- The associated radial gripper parameterizations $\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3]$
- The hypervolume progression $\{\text{HV}^{(0)}, \text{HV}^{(1)}, \dots, \text{HV}^{(T)}\}$
- Total optimization wall-clock time

These solutions represent optimal trade-offs between grasp diversity and triangle area, i.e., grasp robustness and stability.

4.2 Implementation of Two-Finger Gripper Optimization

This chapter presents the optimization framework developed for a two-finger gripper with a spline-based gripping contour. The framework supports both evolutionary multi-objective optimization and Bayesian multi-objective optimization. The evolutionary algorithms are implemented using the `pymoo` library [18], while Bayesian optimization is implemented using the `Ax` platform and its underlying `BoTorch` modeling framework [3, 19].

For both approaches, the evaluation of candidate gripper designs is delegated to an external service through an API. The evaluation service constructs the gripper geometry, voxelizes the gripper and workpieces, and calculates the grasp-quality objectives using a modified

version of Snuggle-Pack[92]. This separation provides a modular architecture in which the optimization logic and the domain-specific grasp evaluation remain decoupled.

The same decision variables, parameter bounds, geometry-generation pipeline, workpieces, orientations, and grasp-quality objectives are used for both optimization approaches. This enables their performance to be compared under a common evaluation budget and evaluation procedure.

4.2.1 Decision Variables and Bounds for Two-Finger Gripper Problem

A candidate gripper is represented by the decision vector

$$\mathbf{x} = [d_{\text{gn}} \quad l_{\text{gn}} \quad h_1 \quad h_2 \quad h_3]^T, \quad (4.41)$$

where the variables are transmitted to the evaluation API in the order

$$\mathbf{x} = [\text{depth_gn}, \text{length_gn}, \text{h_p1_spline}, \text{h_p2_spline}, \text{h_p3_spline}]^T. \quad (4.42)$$

The decision variables and their bounds are summarized in Table 4.3.

Table 4.3: Decision variables of the two-finger gripper optimization problem.

Variable	Configuration parameter	Lower bound	Upper bound
d_{gn}	depth_gn	1	25
l_{gn}	length_gn	1	50
h_1	h_p1_spline	1	15
h_2	h_p2_spline	1	15
h_3	h_p3_spline	1	15

All five decision variables are treated as continuous variables. Candidate values generated by either optimization framework are restricted to the bounded search space

$$\mathcal{X} = [1, 25] \times [1, 50] \times [1, 15]^3. \quad (4.43)$$

For evolutionary optimization, the variables are initially sampled within these bounds and subsequently modified by the operators of the selected algorithm. For Bayesian optimization, Ax initially evaluates the center of the search space and generates additional initialization candidates using Sobol sampling. After sufficient observations have been collected, Ax transitions to model-based candidate generation.

4.2.2 Optimization Framework for Two-Finger Gripper Optimization

The two-finger gripper problem is solved using either an evolutionary multi-objective optimization algorithm or Bayesian multi-objective optimization. Both methods use the same bounded decision space, objective definitions, and external evaluation service.

The common optimization configuration contains:

- Number of objectives n_{obj}
- Decision-variable definitions and bounds
- Optimization direction
- Choice of optimization method
- Evaluation endpoint URL

For evolutionary optimization, the additional configuration parameters include:

- Population size N
- Maximum number of generations G_{max}
- Choice of evolutionary algorithm

For Bayesian optimization, the additional configuration parameters include:

- Maximum number of candidate-generation batches B_{max}
- Maximum number of candidates requested per batch b
- Bayesian generation strategy

In the Bayesian implementation, a batch represents a candidate-generation step rather than an evolutionary generation. The maximum requested number of Bayesian evaluations is therefore

$$E_{\max}^{\text{BO}} = B_{\max} b. \quad (4.44)$$

The actual number of evaluations may be smaller if the generation strategy returns fewer than b candidates in a batch.

Supported Optimization Methods for Two-Finger Gripper Optimization

The framework supports the evolutionary and Bayesian multi-objective optimization methods listed in Table 4.4.

Table 4.4: Supported multi-objective optimization methods for the two-finger gripper.

Family	Identifier	Method
Evolutionary	NSGA2	Non-dominated Sorting Genetic Algorithm II [67]
Evolutionary	NSGA3	Non-dominated Sorting Genetic Algorithm III [68]
Evolutionary	MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [69]
Evolutionary	CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
Evolutionary	SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [70]
Bayesian	AX-MBM	Ax modular BoTorch model for multi-objective Bayesian optimization [3, 19]

For evolutionary algorithms requiring reference directions, such as NSGA-III and MOEA/D, uniform reference directions are generated using the method of Das and Dennis [52]:

$$\Lambda = \text{get_reference_directions}(\text{"das-dennis"}, n_{\text{obj}}, n_{\text{partitions}}). \quad (4.45)$$

The Bayesian implementation uses the Ax generation strategy identified at runtime as `Center+Sobol+MBM:fast`. This strategy consists of the following stages:

1. Evaluation of the center of the bounded decision space

2. Sobol quasi-random sampling for initial space exploration
3. Model-based candidate generation using the Ax modular BoTorch model (MBM)

The Center and Sobol stages do not use a surrogate model or an acquisition function. Once the generation strategy transitions to the MBM stage, Ax uses the available observations to fit probabilistic surrogate models and optimize an acquisition function for multi-objective candidate selection. For continuous multi-objective problems, this commonly involves Gaussian-process surrogates and a hypervolume-improvement acquisition function such as qLogNEHVI.

The exact surrogate and acquisition-function classes are selected by Ax and can depend on the installed Ax and BoTorch versions, the number of observations, and the experiment configuration. Therefore, the active generation stage, surrogate model, and acquisition function should be recorded from the runtime configuration for reproducibility.

4.2.3 Initialization of Two-Finger Gripper Optimization

The initialization process depends on whether evolutionary or Bayesian optimization is selected.

Evolutionary Initialization

The selected evolutionary algorithm initializes a population of N candidate grippers:

$$\mathcal{P}^{(0)} = \{\mathbf{x}_1^{(0)}, \mathbf{x}_2^{(0)}, \dots, \mathbf{x}_N^{(0)}\}, \quad \mathbf{x}_i^{(0)} \in \mathcal{X}. \quad (4.46)$$

Each candidate is sampled within the variable bounds. The complete population is evaluated through the external API before the selected evolutionary algorithm performs its first selection and variation step.

Bayesian Initialization

Bayesian optimization does not maintain an evolutionary population. Instead, it maintains a dataset containing the completed gripper evaluations:

$$\mathcal{D}_t = \{(\mathbf{x}_i, \mathbf{k}(\mathbf{x}_i))\}_{i=1}^t, \quad (4.47)$$

where

$$\mathbf{k}(\mathbf{x}) = \begin{bmatrix} k_1(\mathbf{x}) & k_2(\mathbf{x}) & \cdots & k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^T \quad (4.48)$$

contains the grasp-quality KPIs returned by the evaluation service.

The Ax generation strategy initially evaluates the center of the bounded search space. Additional initialization candidates are generated using Sobol quasi-random sampling. Sobol sampling provides broad and approximately space-filling coverage of the five-dimensional search space before a surrogate model is introduced.

After the initialization requirements of the generation strategy have been satisfied, Ax transitions to the model-based MBM stage. From this point onward, new gripper designs are selected using surrogate predictions and predictive uncertainty.

4.2.4 Objective Function Evaluation of Two-Finger Gripper via API

For every candidate, the decision variables are sent to the evaluation API as an ordered list. The API inserts these values into a complete gripper configuration:

$$\mathbf{x} \longrightarrow \mathbf{c}_{\text{gripper}}(\mathbf{x}) \longrightarrow \mathcal{G}(\mathbf{x}), \quad (4.49)$$

where $\mathbf{c}_{\text{gripper}}$ denotes the gripper configuration and $\mathcal{G}(\mathbf{x})$ the resulting geometry.

The configuration also contains fixed parameters, including the contour type, basic jaw dimensions, jaw angle, stroke, and tool-center-point translation. The spline contour is selected by the static configuration and is not itself an optimization variable. The completed configuration is used to create the jaw geometry using CadQuery [105] which converts it into the mesh representation required for evaluation.

Grasp Evaluation

The generated gripper is evaluated against each configured workpiece. The gripper and workpiece geometries are then voxelized to obtain three-dimensional occupancy grids.

The grasp evaluation uses a modified version of Snuggle-Pack, a three-dimensional packing algorithm that combines FFT-based spatial analysis with proximity heuristics [92]. In the original method, correlations between voxel grids are used to identify collision-free object placements. Here, the same principle is adapted to evaluate relative gripper–workpiece placements.

Let $s_{\mathcal{G}}$ and $s_{\mathcal{W}}$ denote the occupancy grids of the gripper and workpiece. Their spatial correlation produces a collision map:

$$\zeta_{\mathcal{G},\mathcal{W}}(\mathbf{q}) = \mathcal{F}^{-1} \left(\overline{\mathcal{F}(s_{\mathcal{G}})} \mathcal{F}(s_{\mathcal{W}}) \right) (\mathbf{q}), \quad (4.50)$$

where $\mathbf{q}$ is a relative displacement and $\mathcal{F}$ denotes the Fourier transform. Values greater than zero indicate overlapping voxels, whereas zero values identify collision-free relative placements.

The collision map is combined with Gaussian proximity distributions to determine how closely the fingers surround the workpiece at feasible placements. These calculations are performed using FFT-based convolution and GPU-accelerated PyTorch operations. The resulting grasp quality is represented by the KPI `max_prox`. If the required opening cannot be achieved within the available gripper stroke, the KPI is set to zero.

Objective Evaluation

The evaluation service returns one grasp-quality KPI for each evaluated workpiece–orientation combination. Since a larger proximity KPI represents a better grasp result, the natural optimization problem is

$$\max_{\mathbf{x} \in \mathcal{X}} k_m(\mathbf{x}), \quad m = 1, \dots, n_{\text{obj}}. \quad (4.51)$$

The natural objective vector is therefore

$$\mathbf{k}(\mathbf{x}) = \begin{bmatrix} k_1(\mathbf{x}) & k_2(\mathbf{x}) & \cdots & k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^T. \quad (4.52)$$

The objective-direction convention is handled differently by the two optimization frameworks. Since `pymoo` assumes minimization, the evolutionary algorithms negates the objective vector

$$\mathbf{f}_{\text{EA}}(\mathbf{x}) = -\mathbf{k}(\mathbf{x}) = \begin{bmatrix} -k_1(\mathbf{x}) & -k_2(\mathbf{x}) & \cdots & -k_{n_{\text{obj}}}(\mathbf{x}) \end{bmatrix}^T. \quad (4.53)$$

In the Bayesian implementation, the external evaluation service values are left as is, since `Ax` assumes maximization:

$$\mathbf{f}_{\text{BO}}(\mathbf{x}) = \mathbf{k}(\mathbf{x}). \quad (4.54)$$

The maximization direction is specified in the `Ax` optimization configuration. Consequently, the observed KPI values do not need to be negated before they are attached to the corresponding `Ax` trial. This also preserves the physical meaning of the KPI values when the final Pareto frontier is returned.

If n_W workpieces and n_R orientations are evaluated separately, the number of objectives is

$$n_{\text{obj}} = n_W n_R. \quad (4.55)$$

For the currently configured single orientation, each workpiece contributes one objective:

$$n_{\text{obj}} = n_W. \quad (4.56)$$

For performance comparisons, the results from both optimization frameworks are converted to the common natural maximization convention $\mathbf{k}(\mathbf{x})$.

4.2.5 Optimization Loop for Two-Finger Gripper

The external evaluation pipeline is shared by the evolutionary and Bayesian optimization approaches. The methods differ primarily in how they use the completed evaluations to generate subsequent candidates.

For every candidate, the common evaluation steps are:

1. Send the candidate decision vector to the external evaluation API.
2. Map the decision variables to the complete gripper configuration.
3. Generate the spline-based gripper geometry using CadQuery.
4. Voxelize the gripper and configured workpieces.
5. Calculate FFT-based collision maps using the modified Snuggle-Pack procedure.
6. Calculate one proximity KPI for each workpiece–orientation combination.
7. Return the KPI vector to the selected optimization framework.

Evolutionary Optimization Loop

In every evolutionary generation, the selected algorithm generates a population of candidate decision vectors. The candidates are evaluated through the common evaluation pipeline, and the resulting KPI values are negated before being supplied to `pymoo`.

The population update can be expressed abstractly as

$$\mathcal{P}^{(g+1)} = \mathcal{A} \left(\mathcal{P}^{(g)}, \mathbf{F}_{\text{EA}}^{(g)} \right), \quad (4.57)$$

where $\mathcal{A}$ represents the selected evolutionary algorithm and $\mathbf{F}_{\text{EA}}^{(g)}$ contains the minimization-form objective vectors evaluated in generation g .

Depending on the selected algorithm, the next population is produced using operations such as selection, crossover, mutation, decomposition, particle movement, or survival based on hypervolume contribution.

Bayesian Optimization Loop

During the model-based stage, Ax and BoTorch use the completed trial dataset $\mathcal{D}_t$ to fit probabilistic surrogate models:

$$\hat{\mathbf{k}}_t = \mathcal{M}(\mathcal{D}_t), \quad (4.58)$$

where $\mathcal{M}$ denotes the surrogate-modeling procedure.

The surrogate models approximate the relationship between the gripper parameters and the workpiece-specific KPI values. In addition to predicted objective values, they estimate predictive uncertainty in regions containing few or no observations.

An acquisition function $\alpha(\mathbf{x} \mid \mathcal{D}_t)$ uses these predictions to balance:

- Exploitation of gripper designs expected to improve the current Pareto frontier
- Exploration of uncertain regions of the bounded decision space

For a requested batch size b , the next candidate batch is generated as

$$\mathcal{X}_{t+1} = \arg \max_{\substack{\mathcal{B} \subset \mathcal{X} \\ |\mathcal{B}| \leq b}} \alpha(\mathcal{B} \mid \mathcal{D}_t). \quad (4.59)$$

Each generated candidate is evaluated using the external service. The KPI observation is then attached to the corresponding Ax trial:

$$\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{x}_{t+1}, \mathbf{k}(\mathbf{x}_{t+1}))\}. \quad (4.60)$$

After new trials have been completed, the surrogate models are updated using the expanded dataset, and another candidate batch is generated. This process continues until the Bayesian evaluation budget is exhausted.

Although multiple candidates can be requested in one Bayesian batch, the current implementation evaluates the candidates one after another through the external API. The implementation therefore performs *sequential batch evaluation*, rather than parallel batch evaluation.

4.2.6 Termination Criteria for Two-Finger Gripper Optimization

The termination criteria depend on the selected optimization approach.

Table 4.5: Termination criteria for evolutionary and Bayesian optimization.

Approach	Criterion	Symbol	Description
Evolutionary	Design-space tolerance	x_{tol}	Change in decision variables below threshold
Evolutionary	Objective-space tolerance	f_{tol}	Relative objective improvement below threshold
Evolutionary	Maximum generations	G_{max}	Maximum number of generations reached
Evolutionary	Maximum evaluations	E_{max}	Maximum number of evaluations reached
Evolutionary	Hypervolume	HV_{max}	Hypervolume progression stagnates
Bayesian	Maximum batches	B_{max}	Maximum number of candidate batches reached
Bayesian	Maximum evaluations	$E_{\text{max}}^{\text{BO}}$	Bayesian evaluation budget exhausted

For evolutionary optimization, the design-space and objective-space tolerances are evaluated over a sliding window of w generations.

The current Bayesian implementation uses budget-based termination. It performs at most B_{max} candidate-generation batches and requests up to b candidates in each batch. Therefore,

$$E_{\text{max}}^{\text{BO}} = B_{\text{max}}b. \quad (4.61)$$

4.2.7 Performance Monitoring for Two-Finger Gripper

The hypervolume indicator [104] is used to monitor both evolutionary and Bayesian optimization. Before calculating hypervolume, all objective vectors are represented using the same natural maximization convention:

$$\mathbf{k}(\mathbf{x}) = [k_1(\mathbf{x}), \dots, k_{n_{\text{obj}}}(\mathbf{x})]^T. \quad (4.62)$$

For evolutionary optimization, a callback records the optimization state after each generation. Let $\mathcal{K}_{\text{EA}}^{(g)}$ denote the non-dominated KPI vectors available after generation g . The corresponding hypervolume is

$$\text{HV}_{\text{EA}}^{(g)} = \text{HV}(\mathcal{K}_{\text{EA}}^{(g)}, \mathbf{r}). \quad (4.63)$$

For Bayesian optimization, hypervolume can be calculated after every completed trial or after every completed candidate batch. Let $\mathcal{K}_{\text{BO}}^{(t)}$ denote the non-dominated KPI vectors obtained from the first t completed Bayesian trials. The corresponding hypervolume is

$$\text{HV}_{\text{BO}}^{(t)} = \text{HV}(\mathcal{K}_{\text{BO}}^{(t)}, \mathbf{r}). \quad (4.64)$$

Under the maximization convention, the fixed reference point $\mathbf{r}$ must be worse than the objective values of interest. The same reference point and objective scaling must be used for all compared methods. An increasing hypervolume generally indicates improvement in convergence toward the Pareto front, coverage of the objective space, or both.

Evolutionary generations and Bayesian batches can contain different numbers of candidate evaluations. Therefore, the primary comparison is performed using hypervolume as a function of the cumulative number of objective evaluations:

$$\text{HV} = \text{HV}(E), \quad (4.65)$$

where E is the number of completed external API evaluations.

The framework additionally records total wall-clock time because every function evaluation includes API communication, CadQuery geometry generation, voxelization, and GPU-based FFT spatial analysis. Hypervolume as a function of wall-clock time may therefore also be used to compare practical optimization efficiency.

4.2.8 Optional Hyperparameter Optimization for Two-Finger Gripper Problem

For the evolutionary algorithms, the framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-

optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The negative mean hypervolume across the runs is used as the meta-objective:

$$\min_{\theta} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \text{HV}(\mathcal{K}_s, \mathbf{r}) \right\}. \quad (4.66)$$

The best hyperparameter configuration is subsequently applied to the main evolutionary optimization run.

The current Bayesian implementation uses the generation strategy, surrogate configuration, and acquisition-function configuration selected by Ax. A separate meta-optimization of Bayesian model hyperparameters is therefore not currently performed. The user specifies the candidate batch size and optimization budget, while Ax manages the transition from initialization to model-based optimization.

For reproducibility, each Bayesian experiment records:

- Ax and BoTorch software versions
- Random seed, where supported
- Generation strategy
- Number of initialization trials
- Candidate batch size
- Total evaluation budget
- Active surrogate and acquisition-function classes after entering MBM

4.2.9 Output of Two-Finger Gripper Optimization

The optimization produces a set of non-dominated gripper configurations. The Pareto-optimal set is defined as

$$\mathcal{P}^* = \{\mathbf{x} \in \mathcal{P} \mid \nexists \mathbf{x}' \in \mathcal{P} : \mathbf{x}' \prec \mathbf{x}\}, \quad (4.67)$$

where $\mathbf{x}' \prec \mathbf{x}$ means that

$$f_m(\mathbf{x}') \leq f_m(\mathbf{x}) \quad \forall m \quad (4.68)$$

and

$$f_j(\mathbf{x}') < f_j(\mathbf{x}) \quad \text{for at least one objective } j. \quad (4.69)$$

The final output includes:

- The Pareto-optimal decision vectors $\mathbf{x} = [d_{\text{gn}}, l_{\text{gn}}, h_1, h_2, h_3]^T$
- The corresponding objective KPI values
- The hypervolume progression $\{\text{HV}^{(0)}, \text{HV}^{(1)}, \dots, \text{HV}^{(T)}\}$
- Total optimization wall-clock time

The non-dominated solutions represent the best trade-offs between grasp quality for the configured workpieces and orientations.

5 Validation

5.1 Validation of Results for Suction-Cup Gripper

5.2 Validation of Results for Two-Finger Gripper

6 Evaluation

6.1 Results for Suction-Cup Gripper

6.1.1 Results for Evolutionary Algorithms

6.1.2 Results for Bayesian Algorithms

6.2 Results for Two-Finger Gripper

6.2.1 Results for Evolutionary Algorithms

6.2.2 Results for Bayesian Algorithms

7 Summary

7.1 Conclusion

7.2 Limitations

7.3 Future Work

A Appendix

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

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